

Lecture 25: Application of the NP Lemma

The NP lemma is used in the construction of Most Powerful (MP) and Uniformly Most Powerful (UMP) tests.

MP size- α test ϕ_0 is an MP size- α test for testing

$H_0: \theta = \theta_0$ (or $\theta \in \Theta_0$) against $H_A: \theta = \theta_1$, if

$$(i) E_{\theta_0} \phi_0(X) \leq \alpha \quad (\text{or } \sup_{\theta \in \Theta_0} E \phi_0(X) \leq \alpha) \quad (\text{size condition})$$

and (ii) $E_{\theta_1} \phi_0(X) > E_{\theta_1} \phi(X) \quad \forall$ test ϕ satisfying (i) (power condition)

(ϕ_0 is the MP size- α test among all size- α tests)

UMP size- α test ϕ_0 is UMP size- α test for testing $H_0: \theta = \theta_0$ (or $\theta \in \Theta_0$) against $H_A: \theta \in \Theta_A$ if

$$(i) E_{\theta_0} \phi_0(X) \leq \alpha \quad (\text{or } \sup_{\theta \in \Theta_0} E \phi_0(X) \leq \alpha) \quad (\text{size})$$

$$(ii) E_{\theta} \phi_0(X) \geq E_{\theta} \phi(X) \quad \forall \theta \in \Theta_A \text{ and } \forall \phi \text{ satisfying (i)} \quad (\text{power})$$

(ϕ_0 is the UMP (size- α) $\forall \theta \in \Theta_A$ test among all size- α tests)

The NP lemma is applied essentially to construct the MP size- α test for a simple null $H_0: \theta = \theta_0$ ag. a simple alternative $H_A: \theta = \theta_1$.
Simple extension/discussion/condition will lead to the construction of tests with composite null and/or alternative hypotheses.

Use of NP lemma for an MP size- α test (continuous distn case, replace ' $\leq$ ' by ' $\leq$ ' in size condn & integrals by summation in discrete distn case)

Consider a simple H_0 ag. a simple H_A for a contin. prob. model.

let f_0 be the density under H_0 , f_A under H_A .

To construct the MP size- α test, we're looking for a region w_0 (the critical region)

$$(i) \int_{w_0} f_0 dx = \alpha$$

and (ii) $\int_{w_0} f_A dx$ is max^m among all w satisfying (i) (w , including w_0 , $\subset \mathcal{X}$)

In the NP lemma, take $g_0 = f_A$, $g_1 = f_0$, $c_1 = \alpha$ (fixed constant, in $(0,1)$),
 $k_1 = k$ (some constant to be evaluated by the size-condition)
 $k_2 = k_3 = \dots = 0$.

Then, by the lemma, $f_A \geq k f_0$ in ω_0
 $<$ outside ω_0 (in $\mathcal{X} - \omega_0$)

$$\text{and } \int_{\omega_0} f_A dx \geq \int_{\omega_0} k f_0 dx \quad \forall \omega \ni \int_{\omega} f_0 dx = \alpha; \text{ i.e.,}$$

$$\omega_0 \text{ maximizes } \int_{\omega} f_A dx \quad \forall \omega \in \mathcal{X} \ni \int_{\omega} f_0 dx = \alpha \quad (\omega_0 \text{ maximizes power given size-}\alpha)$$

In fact, we would consider the ratio $\frac{f_A(x)}{f_0(x)}$, leading to a statistic $T(x)$

$$\frac{f_A(x)}{f_0(x)} = T(x) \geq k \quad \text{within } \omega_0.$$

The cut-off value k will be obtained from the size-condition, i.e.,
 k is $\ni P_{H_0}(T(x) \geq k) = \alpha$

(Spl. cases: (i) $\alpha = 0$, never reject H_0 , $k = \alpha$
(ii) $\alpha = 1$, always reject H_0 , $k = 0$)

Example 1. A single observation from a population with pdf
 $f_0(x) = 2x^{2-1}$, $0 < x < 1$, $0 > 0$ is considered.
Construct the MP size- α test for $H_0: \theta = 3$ ag. $H_A: \theta = 2$.

Solns $\omega_0: \frac{f_A(x)}{f_0(x)} = \frac{2x}{3x^2} \geq k$ i.e., $x \leq k^*$, $k^* \ni P_{H_0}(X \leq k^*) = \alpha$
i.e., $\int_0^{k^*} 3x^2 dx = \alpha \Rightarrow k^* = (\alpha)^{1/3}$. If $\alpha = 0.05$ (fixed at this value),
then $k^* = 0.368$.

(If the single sample observation is ≤ 0.368 , reject H_0 at 5% level of significance, or, do not reject H_0 at this level).

What is the power of this MP size- α test?

$$\text{Power} = P_{H_A}(X \leq k^*) = \int_0^{0.368} 2x dx = (0.368)^2 = 0.135.$$

(Why does the MP test has such a low power?
How can you improve it?)